

Week 5 Assignment 2: Configuring Site-to-Site VPNs

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Introduction

I engaged in configuring and monitoring an IPSec VPN tunnel between two routers using Cisco IOS. The goal was to ensure that traffic between the LAN on R1 and the LAN on R3 was securely encrypted using the IPSec protocol, triggered by interesting traffic defined through access control lists. As part of the validation process, I used the show crypto ipsec sa command to observe the state of the tunnel, examine packet encryption and decryption statistics, and verify the active Security Associations (SAs). This exercise provided practical insight into how IPSec tunnels are established, maintained, and verified in a live environment.

Part 1

Enabling Security Features

To begin configuring the Site-to-Site VPN, I first needed to ensure that the security features required for VPN functionality were available on Router R1. I accessed R1 through the CLI in Cisco Packet Tracer and entered privileged EXEC mode using the enable command. To check if the required security package was active, I issued the show version command and examined the “Technology Package License Information” section. I noticed that the securityk9 module was not active, as it was listed as “None.”

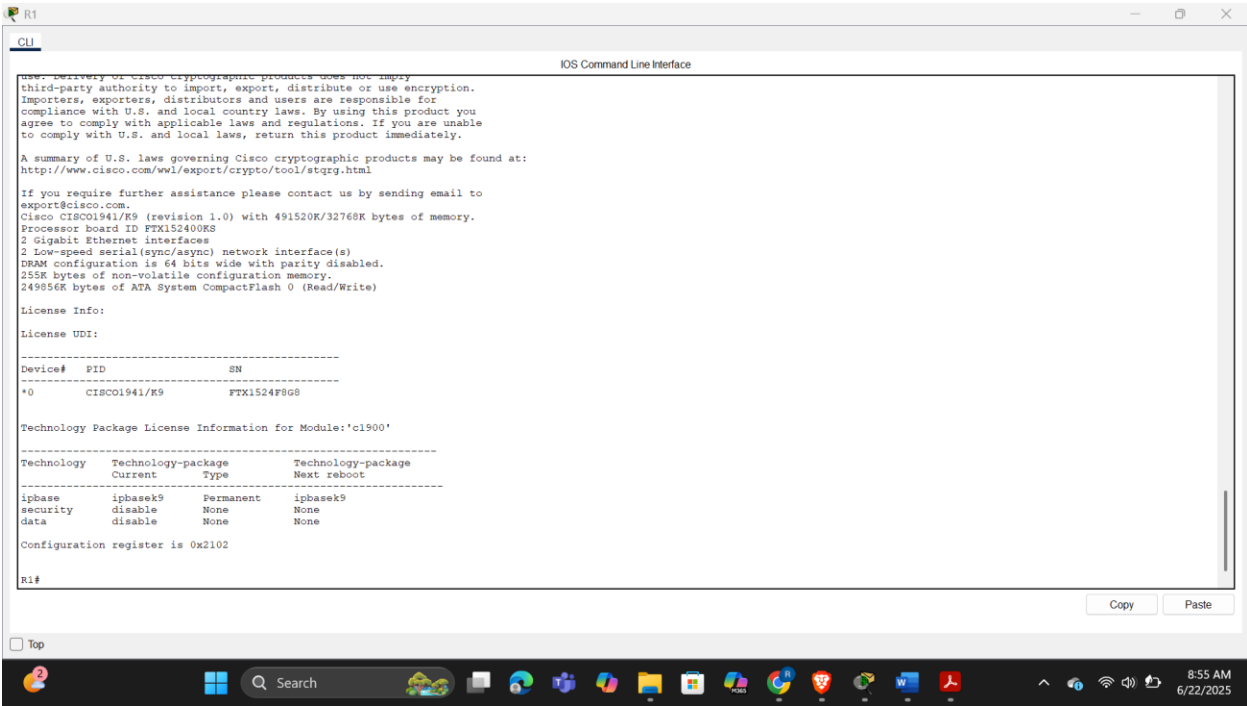
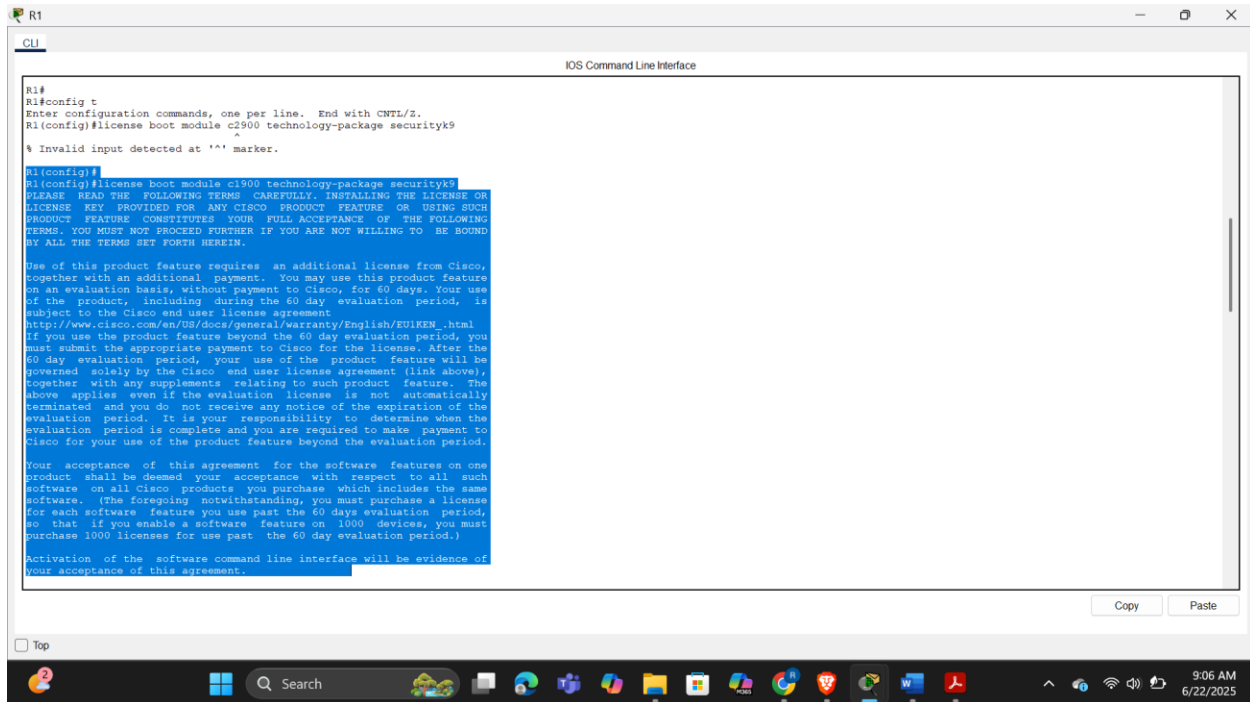


Fig 1.0 verifying security parameters on R1 using show version command

To activate it, I entered global configuration mode with configure terminal and executed the command `license boot module c1900 technology-package securityk9` to instruct the router to enable the security package at the next boot. After that, I exited configuration mode by typing `end` and saved the current running configuration to the startup configuration using `copy running-config startup-config` to ensure the license change would persist after reboot.



```
R1#
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#license boot module c1900 technology-package securityk9
^
% Invalid input detected at '^' marker.

R1(config)#
R1(config)#license boot module c1900 technology-package securityk9
PLEASE READ THE FOLLOWING TERMS CAREFULLY. INSTALLING THE LICENSE OR
LICENSE KEY PROVIDED FOR ANY CISCO PRODUCT FEATURE OR USING SUCH
PRODUCT FEATURE CONSTITUTES YOUR FULL ACCEPTANCE OF THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

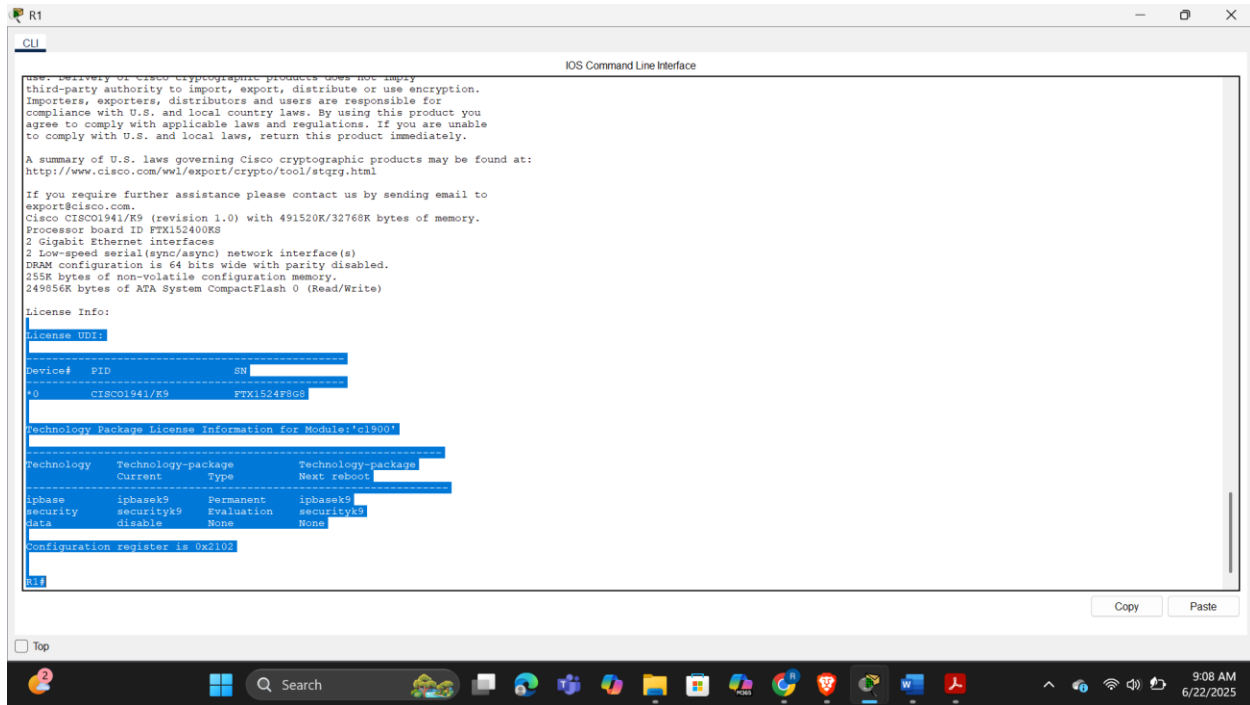
Use of this product feature requires an additional license from Cisco,
together with an additional payment. You may use this product feature
on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the product, including during the 60 day evaluation period, is
subject to the Cisco and user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day evaluation period, your use of the product feature will be
governed solely by the Cisco and user license agreement (link above),
together with any supplements relating to such product feature. The
above applies even if the evaluation license is not automatically
terminated and you do not receive any notice of the expiration of the
evaluation period. It is your responsibility to determine when the
evaluation period is complete and you are required to make payment to
Cisco for your use of the product feature beyond the evaluation period.

Your acceptance of this agreement for the software features on one
product shall be deemed your acceptance with respect to all such
software on all Cisco products you purchase, which includes the same
software. (The foregoing notwithstanding, you must purchase a license
for each software feature you use past the 60 days evaluation period,
so that if you enable a software feature on 1000 devices, you must
purchase 1000 licenses for use past the 60 day evaluation period.)

Activation of the software command line interface will be evidence of
your acceptance of this agreement.
```

Fig 1.1 activating k9 security module on R1

I then reloaded the router with the `reload` command and confirmed the prompt to restart. Once the router rebooted, I again used the `show version` command to confirm the activation. This time, the output showed that the `securityk9` technology package was in evaluation mode and successfully activated, confirming that Router R1 was now equipped with the necessary security capabilities to proceed with VPN configuration. Once the router rebooted, I verified that the security technology package had been successfully enabled by running the `show version` command. In the output, I looked for the “`securityk9`” feature and confirmed that its status was “`Enabled`.” This indicated that the router now had the cryptographic capabilities required to configure IPsec VPNs.



```
user: Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wvl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco IOS01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400K8
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:
License UDI:
-----
Device# PID SN
-----
*0 CISC01941/K9 FTX1524F8G8

Technology Package License Information for Module:'C1900'
-----
Technology Technology-package Type Technology-package
Current Next reboot
-----
ipbase ipbasek9 Permanent ipbasek9
security securityk9 Evaluation securityk9
data disable None None
Configuration register is 0x2102
```

Fig 1.2 verifying the enabled K9 security on R1

Part 2

Configuring IPsec Parameters on R1

To verify baseline connectivity during the IPsec configuration process on R1, I initiated a ping from PC-A to PC-B. Since both devices reside within the same local network and are not part of the defined "interesting traffic" for the VPN, this test confirmed that local communication was functioning properly without encryption. The successful ping indicated that IP addressing, static routing, and interface configurations were correctly set up and operational. This step ensured that the network was behaving as expected prior to enabling IPsec features.

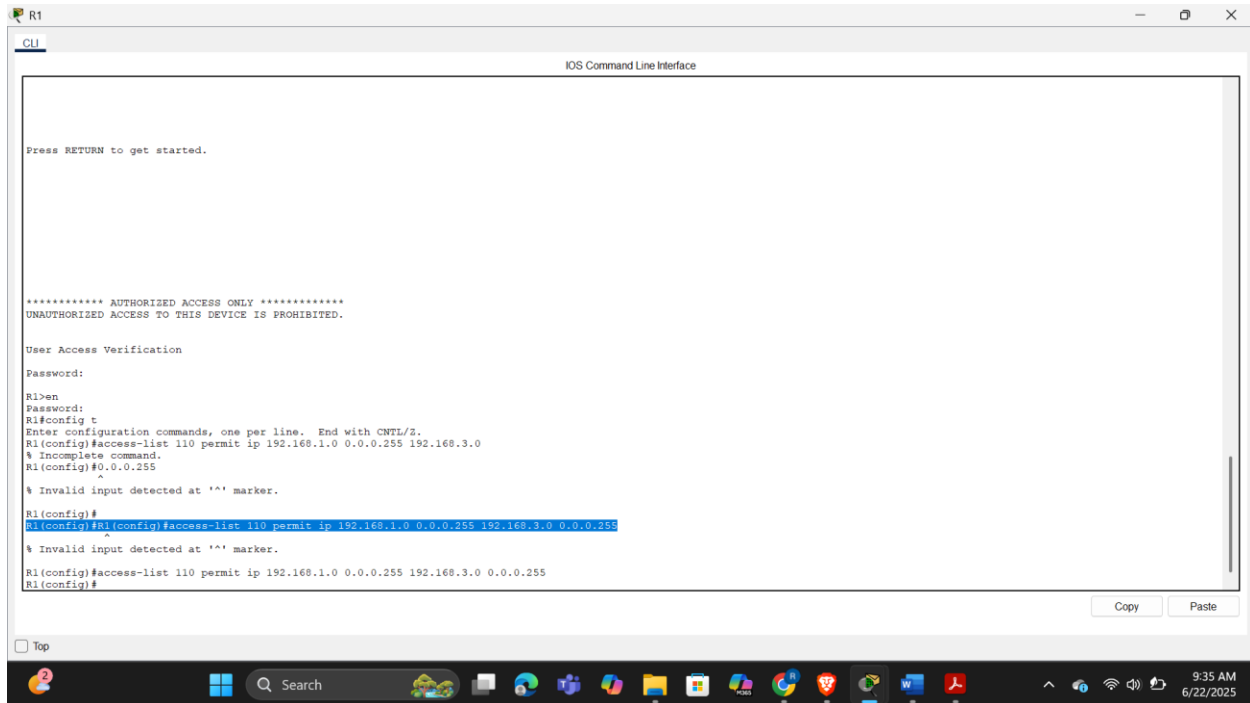


Fig 1.4 identifying interesting traffic

Configuring IKE Phase 1 (ISAKMP Policy)

Next, I set up the parameters for IKE Phase 1, which is responsible for establishing a secure tunnel and negotiating the initial encryption parameters. I created an ISAKMP policy using `crypto isakmp policy 10`. Inside the policy configuration mode, I specified the encryption method as AES-256 (`encryption aes 256`), set the authentication method to pre-shared key (authentication pre-share), and selected Diffie-Hellman group 5 (group 5) for key exchange. These settings ensure that the tunnel setup is secure and the keys are exchanged confidentially. After exiting ISAKMP configuration mode, I configured the pre-shared key `vpnpa55` to be associated with R3's public IP (10.2.2.2) using `crypto isakmp key vpnpa55 address 10.2.2.2`.

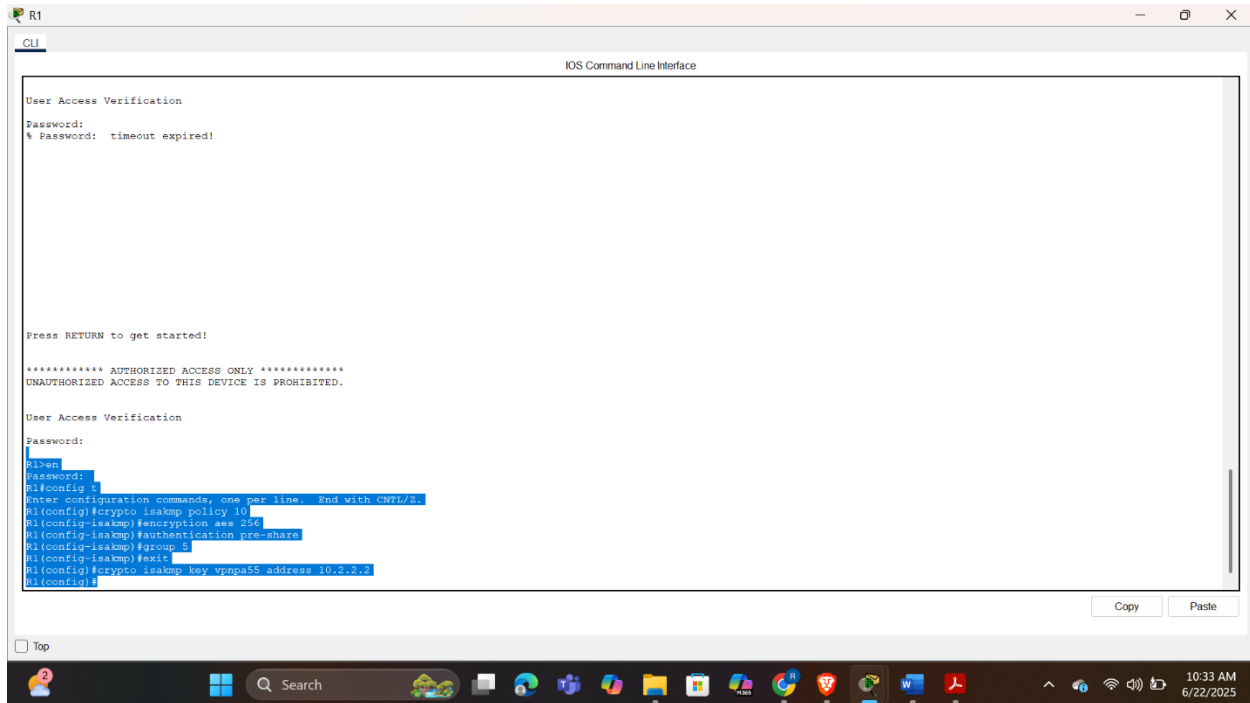


Fig 1.5 Configuring the IKE Phase 1 ISAKMP policy on R1

Configuring IKE Phase 2 (IPsec Policy)

With Phase 1 configured, I moved on to Phase 2, where the actual data encryption settings are defined. I began by creating a transform set named VPN-SET using `crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac`, which specified that the tunnel should use ESP for both encryption (AES) and authentication (SHA-HMAC). Then, I created a crypto map called VPN-MAP using `crypto map VPN-MAP 10 ipsec-isakmp`. Inside the crypto map configuration, I added a description to clarify its purpose (description VPN connection to R3), defined the remote peer (set peer 10.2.2.2), linked the transform set (set transform-set VPN-SET), and matched the traffic defined in ACL 110 (match address 110). This crypto map brings together all the components necessary to establish and maintain the secure IPsec tunnel.

```
R1
CLI
IOS Command Line Interface

Press RETURN to get started!

***** AUTHORIZED ACCESS ONLY *****
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED.

User Access Verification
Password:

R1>en
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#crypto isakmp key vpnps5 address 10.2.3.2
R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
A NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R1(config-crypto-map)#description VPN connection to R3
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#
```

Fig 1.6 Configuring the IKE Phase 2 IPsec policy on R1

Applying the Crypto Map to the Outgoing Interface

Finally, I applied the crypto map to the router's outgoing interface, which is Serial 0/0/0. I entered the interface configuration mode with interface s0/0/0 and then applied the map using crypto map VPN-MAP. This binding ensures that any traffic exiting this interface that matches ACL 110 will trigger the IPsec VPN tunnel to be negotiated and encrypted accordingly.


```
Press RETURN to get started!

***** AUTHORIZED ACCESS ONLY *****
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED.

User Access Verification

Password:

R1>en
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#crypto isakmp key vpnpa55 address 10.2.2.2
R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R1(config-crypto-map)#description VPN connection to R3
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#
R1(config)#R1(config)# interface s0/0/0
^
% Invalid input detected at '^' marker.

R1(config)#interface s0/0/0
R1(config-if)# crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO=6-ISAAMP ON OFF: ISAAMP is ON
R1(config-if)#
```

Fig 1.7 Configuring the crypto map on the outgoing interface

Part 3

Configure IPsec Parameters on R3

To configure the second endpoint of the site-to-site VPN tunnel, I began with R3 by first confirming whether the Security Technology package necessary for VPN support was already enabled. I did this using the show version command. This output revealed whether the securityk9 package was activated. If it had not been active, I would have enabled it using the command license boot module c1900 technology-package securityk9, followed by accepting the end-user agreement, saving the configuration, and reloading the router to activate the security package. Since enabling this package is essential for IPsec functionality, I ensured it was fully operational before proceeding.

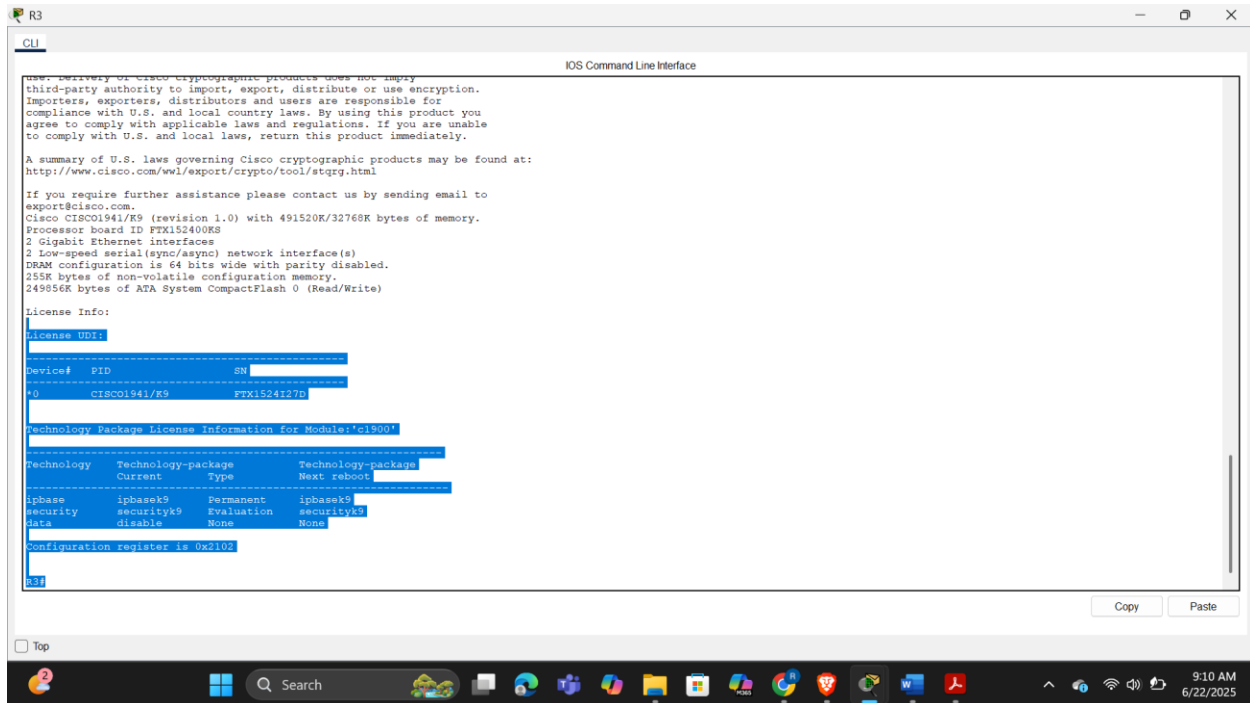


Fig 1.8 verifying security parameters on R3

Configuring router R3 to support a site-to-site VPN with R1

Next, I created an access control list (ACL) on R3 to define interesting traffic that would trigger the VPN tunnel. The ACL was configured using the command `access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255`, which specifies that only traffic originating from R3's LAN subnet and destined for R1's LAN should initiate the VPN tunnel. This ensures encryption is reserved only for relevant communication, maintaining both performance and security.

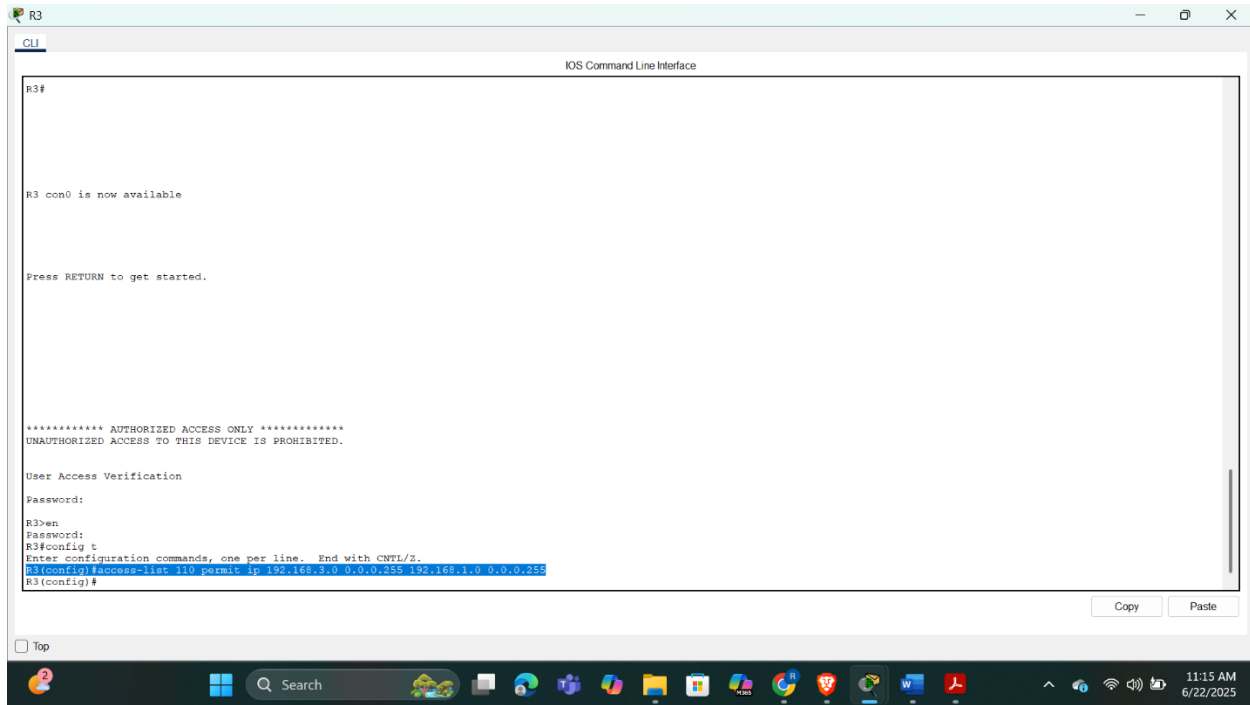
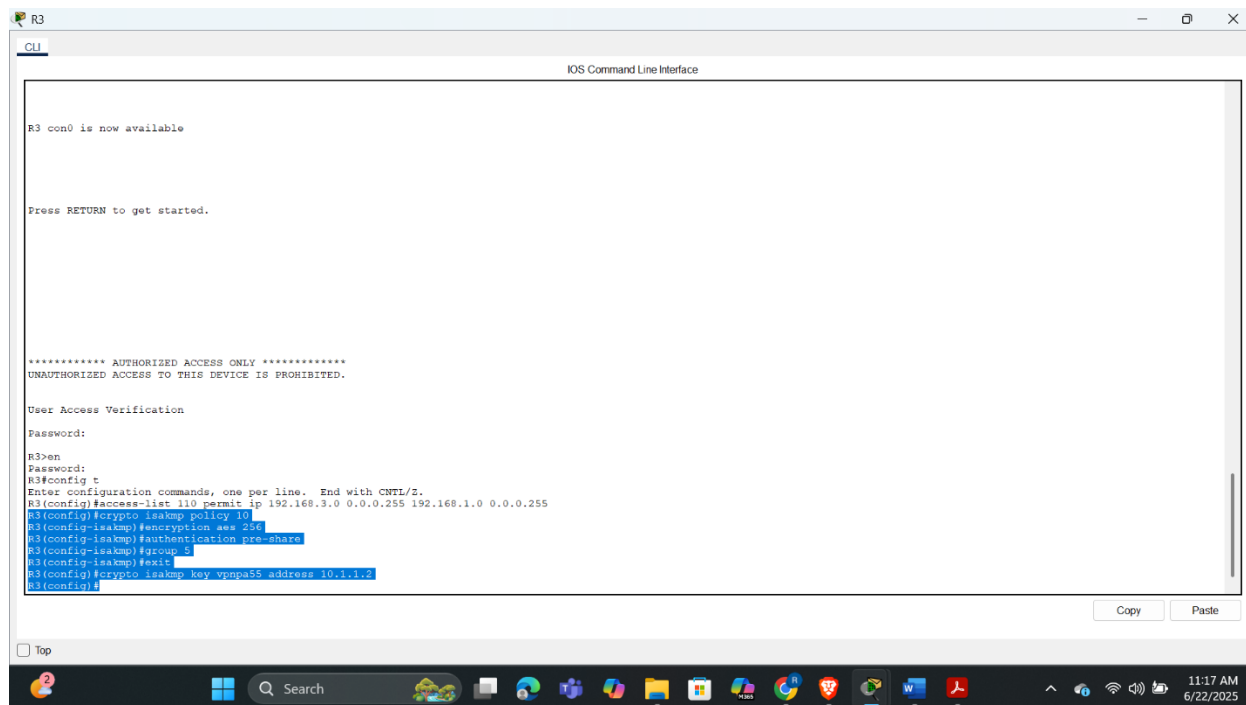


Fig 1.9 Configuring router R3 to support a site-to-site VPN with R1

Configuring the IKE Phase 1 SAKMP properties on R3

With interesting traffic identified, I moved on to setting up the Internet Key Exchange (IKE) Phase 1 policy on R3. I configured policy number 10 with the command `crypto isakmp policy 10`. Inside this policy, I set the encryption method to AES 256-bit with encryption `aes 256`, the authentication method to pre-shared keys using authentication `pre-share`, and specified Diffie-Hellman Group 5 for key exchange via group 5. After exiting the policy configuration, I established the shared key `vpnpa55` to be used specifically with R1's WAN interface by entering `crypto isakmp key vpnpa55 address 10.1.1.2`. This established a secure and consistent key for both ends of the VPN tunnel.



```
R3
CLI
IOS Command Line Interface

R3 con0 is now available

Press RETURN to get started.

***** AUTHORIZED ACCESS ONLY *****
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED.

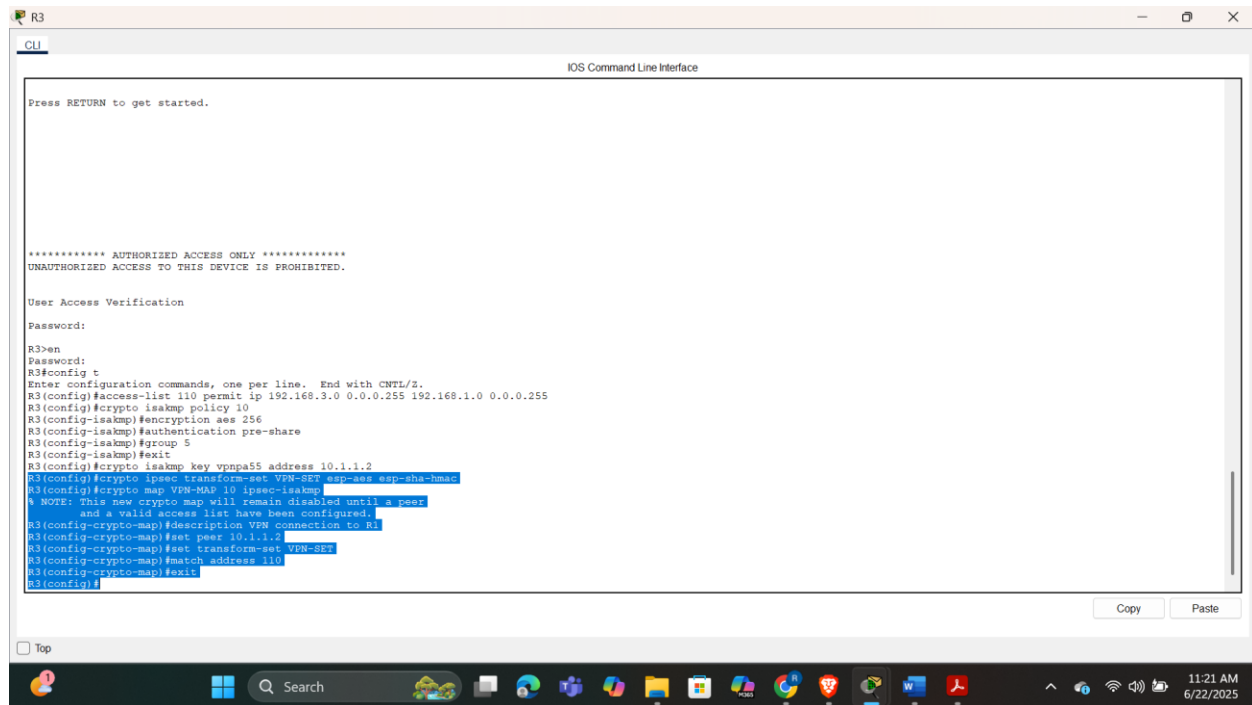
User Access Verification
Password:

R3>en
Password:
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
R3(config)#crypto isakmp policy 10
R3(config-isakmp)#encryption aes 256
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#group 1
R3(config-isakmp)#exit
R3(config)#crypto isakmp key vpnps5 address 10.1.1.2
R3(config)#
```

Fig 2.0 Configuring the IKE Phase 1 SAKMP properties on R3.

Configuring the IKE Phase 2 IPsec policy on R3

Following Phase 1 setup, I configured the IKE Phase 2 IPsec policy. I created a transform set named VPN-SET with the command `crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac`, selecting encryption via AES and hashing via SHA for robust confidentiality and integrity. Then, I created the crypto map named VPN-MAP with sequence number 10 and designated it as an ipsec-isakmp map using `crypto map VPN-MAP 10 ipsec-isakmp`. Within this map, I added a description for clarity, then set R1's IP address as the peer using `set peer 10.1.1.2`. I applied the previously defined transform set and matched the crypto map to ACL 110 using `match address 110`. These settings tie together the parameters necessary for Phase 2 and define how encrypted traffic should be handled.



```
R3
CLI
IOS Command Line Interface

Press RETURN to get started.

***** AUTHORIZED ACCESS ONLY *****
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED.

User Access Verification

Password:
R3>en
Password:
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
R3(config)#crypto isakmp policy 10
R3(config-isakmp)#encryption aes 256
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#group 5
R3(config-isakmp)#exit
R3(config)#crypto isakmp key vpnpa55 address 10.1.1.2
R3(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R3(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R3(config-crypto-map)#description VPN connection to R1
R3(config-crypto-map)#set peer 10.1.1.2
R3(config-crypto-map)#set transform-set VPN-SET
R3(config-crypto-map)#match address 110
R3(config-crypto-map)#exit
R3(config)#
```

Fig 2.1 Configuring the IKE Phase 2 IPsec policy on R3.

Configuring the crypto map on the outgoing interface

Finally, I applied the crypto map to R3's outgoing interface by navigating to the serial interface connected to R1 using interface s0/0/1, and then I bound the map with the command `crypto map VPN-MAP`. This step finalized the IPsec VPN configuration on R3 and enabled the router to actively participate in encrypted communication with R1 once interesting traffic is detected

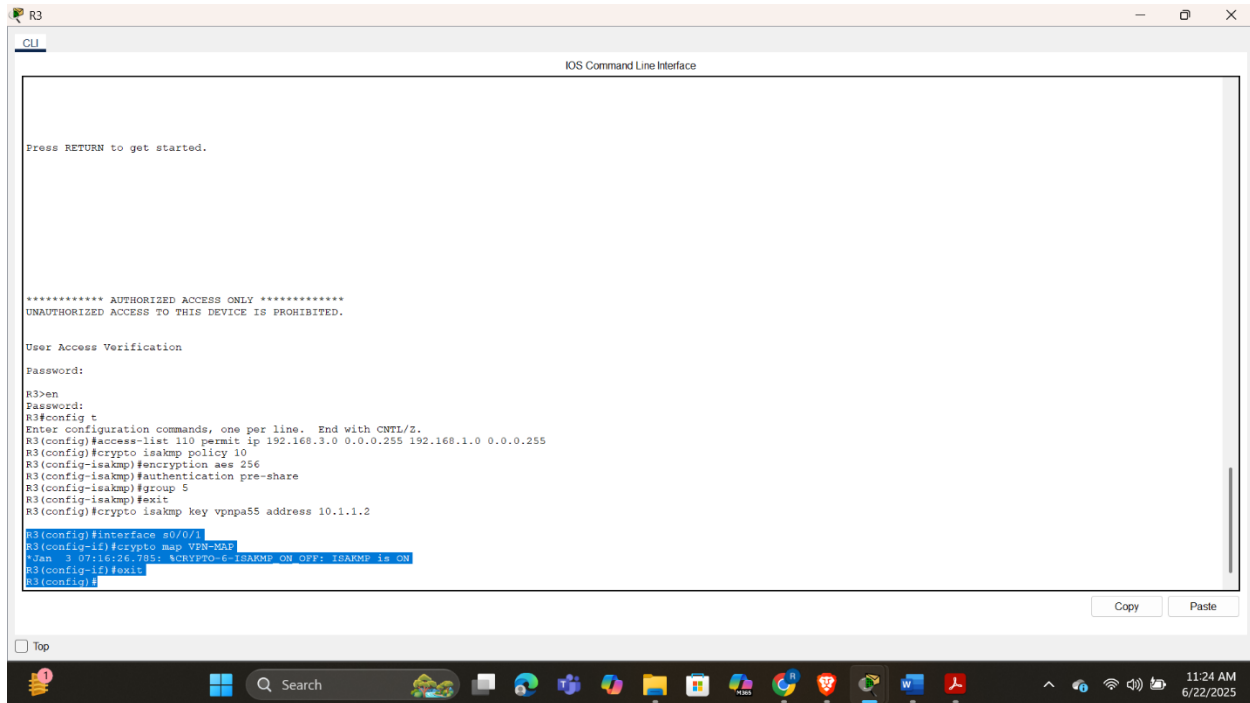


Fig 2.2 Configuring the crypto map on the outgoing interface.

Part 4

Verifying the IPsec VPN

Checking current security statistics

To begin verifying the IPsec tunnel functionality, I first ensured that the tunnel was not active by checking the current security association statistics. On **R1**, I issued the `show crypto ipsec sa` command prior to initiating any interesting traffic. As expected, all packet counters — including `#pkts encaps`, `#pkts encrypt`, `#pkts decaps`, and `#pkts decrypt` — were at **0**, confirming that no traffic had yet triggered the tunnel, and it was currently inactive.

```
R1
CLI
IOS Command Line Interface

UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED

User Access Verification

Password:
Password:
R1>en
Password:
R1#show crypto ipsec sa
interface: Serial0/0/0
Crypto map tag: VPN-MAH, local addr 10.1.1.2

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
current peer 10.2.2.2 port 500
PERMIT, flags=(origin_is_acl,):
  #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
  #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 10.1.1.2, remote crypto endpt.: 10.2.2.2
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
current outbound spi: 0x0(0)

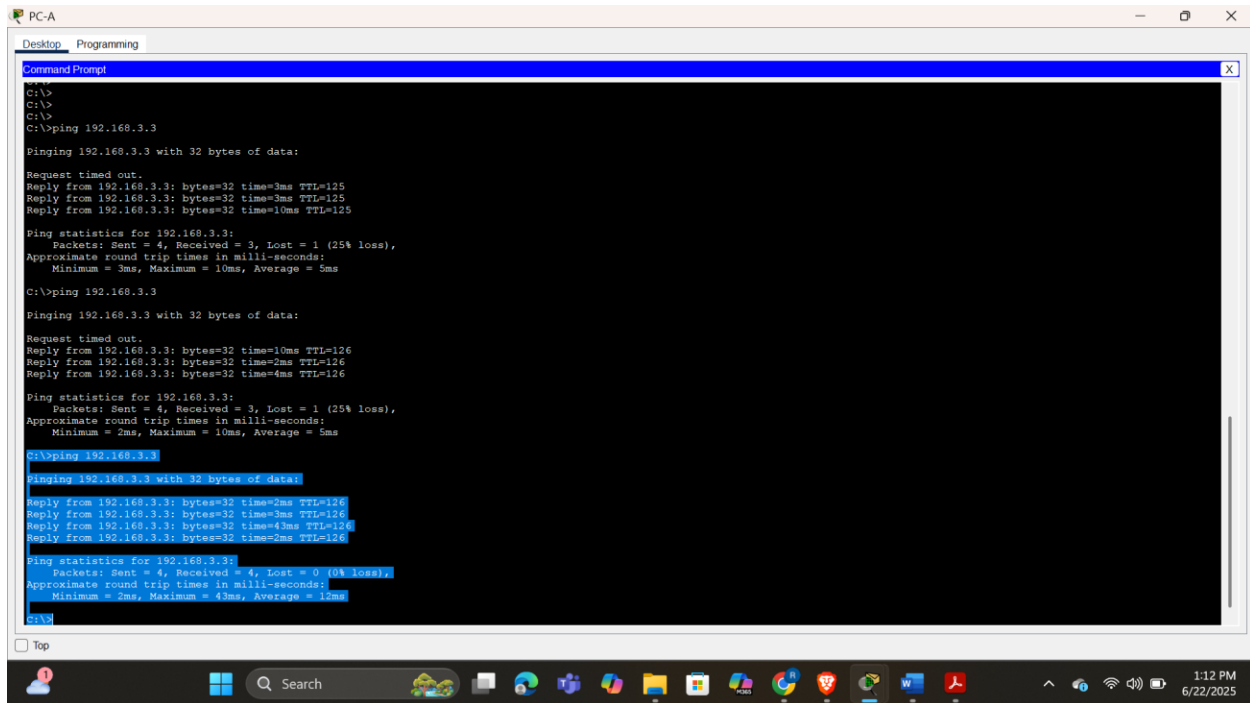
inbound esp sas:
inbound ah sas:
inbound pcp sas:
outbound esp sas:
outbound ah sas:
outbound pcp sas:

R1#
```

Fig 2.3 Tunnel before interesting traffic

Generating interesting traffic

Next, to generate interesting traffic, I moved to **PC-A** (within R1's LAN at 192.168.1.0/24) and pinged **PC-C** (in R3's LAN at 192.168.3.0/24) using the command `ping 192.168.3.3`. Since this traffic matched the ACL 110 configuration identifying interesting traffic, it was expected to trigger the creation of the IPsec tunnel. The first ping attempt timed out, which is normal due to the time needed to negotiate ISAKMP and IPsec phases. However, the subsequent pings were successful, indicating the tunnel had been established.



```
C:\>
C:\>
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time=3ms TTL=125
Reply from 192.168.3.3: bytes=32 time=3ms TTL=125
Reply from 192.168.3.3: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 10ms, Average = 5ms

C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time=10ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
Reply from 192.168.3.3: bytes=32 time=4ms TTL=126

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 10ms, Average = 5ms

C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
Reply from 192.168.3.3: bytes=32 time=3ms TTL=126
Reply from 192.168.3.3: bytes=32 time=43ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126

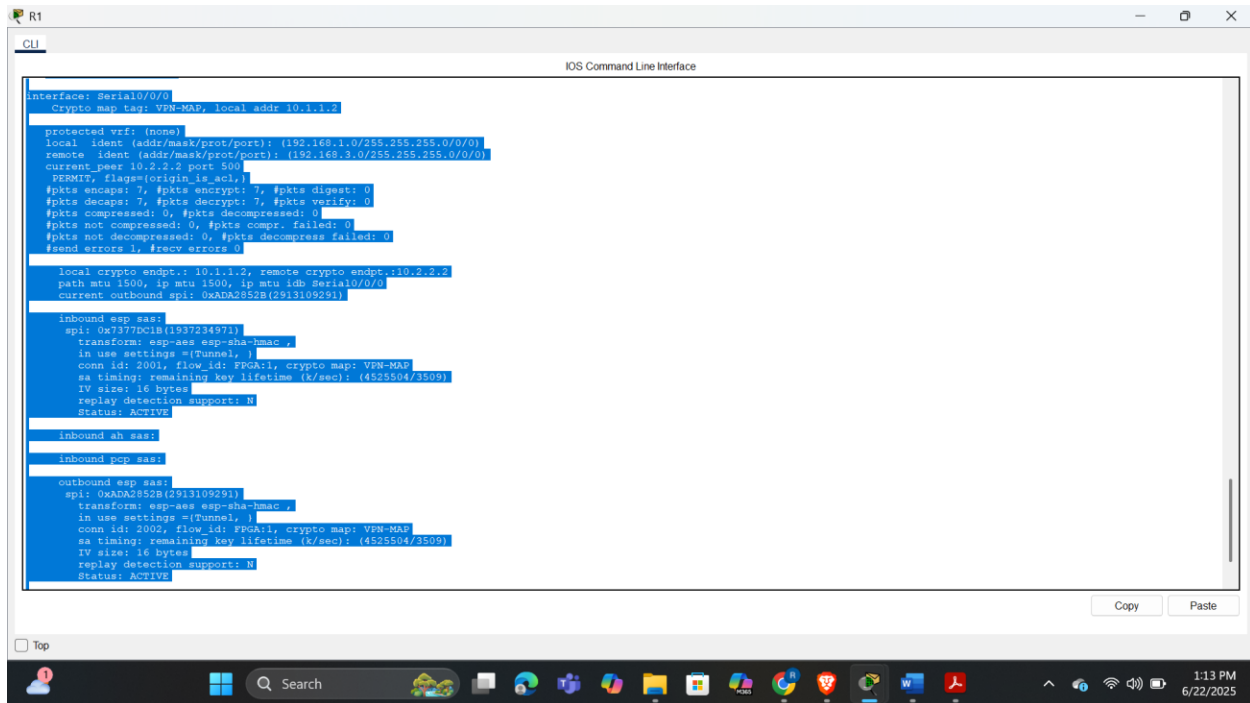
Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 43ms, Average = 13ms

C:\>
```

Fig 2.4 Pinging pc C from Pc B

Checking current security statistics after interesting traffic

Returning to **R1**, I re-issued the `show crypto ipsec sa` command. This time, the output showed non-zero packet counters across the encapsulated, encrypted, decapsulated, and decrypted fields. This confirmed that the IPsec tunnel was now **up and actively processing traffic**, successfully encrypting and decrypting data between the two LANs. The same verification on **R3** yielded matching results, ensuring both ends were synchronized and secure.



```
interface Serial0/0/0
  crypto map tag: VPN-MAP, local addr 10.1.1.3

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
current_peer 10.2.2.2 port 500
  PERMIT, flags={origin_is_acl:}
  #pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
  #pkts decaps: 7, #pkts decrypt: 7, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors: 0, #recv errors: 0

local crypto endpt.: 10.1.1.3, remote crypto endpt.: 10.2.2.2
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
current outbound spi: 0xADA2852B(291310929)

inbound esp sas:
  spi: 0x7377DC1B(193723497)
  transform: esp-aes esp-sha-hmac
  in use settings ={tunnel, }
  conn id: 2001, flow id: FPGA:1, crypto map: VPN-MAP
  sa timing: remaining key lifetime (k/sec): (4525504/3509)
  IV size: 16 bytes
  replay detection support: N
  Status: ACTIVE

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0xADA2852B(291310929)
  transform: esp-aes esp-sha-hmac
  in use settings ={tunnel, }
  conn id: 2002, flow id: FPGA:1, crypto map: VPN-MAP
  sa timing: remaining key lifetime (k/sec): (4525504/3509)
  IV size: 16 bytes
  replay detection support: N
  Status: ACTIVE
```

Fig 2.5 Verifying tunnel after interesting traffic

Generating uninteresting traffic

To further validate the VPN's behavior, I then tested how it handles uninteresting traffic. From **PC-A**, I pinged **PC-B** (also on R1's local LAN). Since this traffic does not match the ACL and is local to R1's network, it was not defined as interesting. After the ping completed successfully, I went back to **R1** and re-checked the IPsec statistics with `show crypto ipsec sa`. As expected, the packet counters had **not changed** since the last interesting traffic ping, confirming that **non-interesting traffic is not being encrypted** and does not engage the VPN tunnel.

```
PC-A
Desktop Programming
Command Prompt
Minimum = 3ms, Maximum = 10ms, Average = 5ms
C:\>ping 192.168.3.3
Pinging 192.168.3.3 with 32 bytes of data:
Request timed out.
Reply from 192.168.3.3: bytes=32 time=10ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
Reply from 192.168.3.3: bytes=32 time=4ms TTL=126
Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 10ms, Average = 5ms
C:\>ping 192.168.3.3
Pinging 192.168.3.3 with 32 bytes of data:
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
Reply from 192.168.3.3: bytes=32 time=3ms TTL=126
Reply from 192.168.3.3: bytes=32 time=43ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 43ms, Average = 12ms
C:\>
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time=46ms TTL=126
Reply from 192.168.2.3: bytes=32 time=14ms TTL=126
Reply from 192.168.2.3: bytes=32 time=15ms TTL=126
Reply from 192.168.2.3: bytes=32 time=2ms TTL=126
Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 46ms, Average = 19ms
C:\>
```

Fig 2.6 Pinging pc B from pc A

```
R1
CLI
IOS Command Line Interface
R1#show crypto ipsec sa
Interface: Serial0/0/0
  Crypto map tag: VPN-MAP, local addr: 10.1.1.3
protected vrf: (none)
  local ident (addr/mask/prot/port): (192.168.1.0/255.255.0.0/0)
  remote ident (addr/mask/prot/port): (192.168.3.0/255.255.0.0/0)
  current peer 10.2.2.2 port 500
    PERMIT, flags=(origin_is_acl),
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
  local crypto endpt.: 10.1.1.2, remote crypto endpt.: 10.2.2.2
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
  current outbound spi: 0xADA2852B(2913109291)
inbound esp sas:
  spi: 0x7373C1B(1937234971)
    transform: esp-aes esp-sha-hmac ,
    in use settings =(Tunnel, )
    conn id: 2001, flow_id: FPGA11, crypto map: VPN-MAP
    sa timing: remaining key lifetime (k/sec): (4525504/3192)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE
inbound ah sas:
inbound pcp sas:
outbound esp sas:
  spi: 0xADA2852B(2913109291)
    transform: esp-aes esp-sha-hmac ,
    in use settings =(Tunnel, )
    conn id: 2002, flow_id: FPGA11, crypto map: VPN-MAP
    sa timing: remaining key lifetime (k/sec): (4525504/3192)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE
```

Fig 2.7 tunnel showing No change since there is no interesting traffic

With these observations and command outputs, I concluded that the IPsec Site-to-Site VPN between **R1** and **R3** is fully operational. The tunnel is dynamically established only when interesting traffic is detected, encrypts and decrypts packets as expected, and safely ignores non-

interesting traffic. This successful end-to-end verification demonstrates that the configuration meets the intended security objectives.

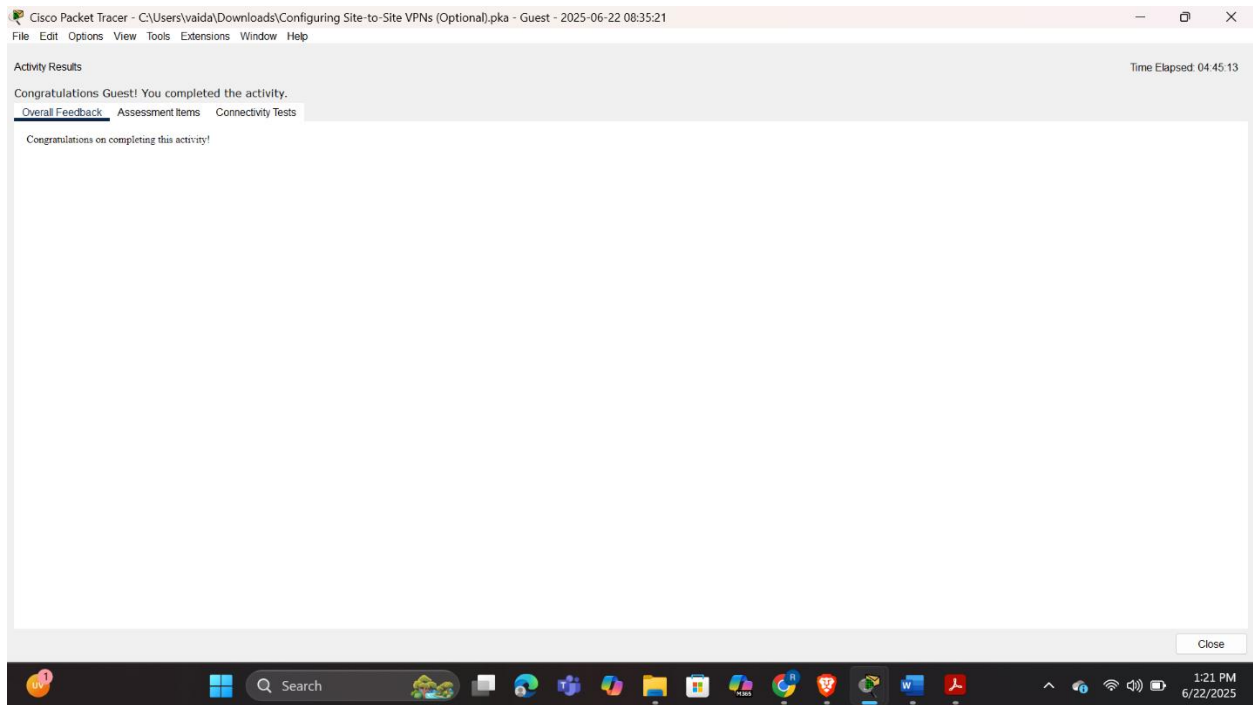


Fig 2.8 completion status

Conclusion

Through this exercise, I successfully configured and tested an IPSec VPN tunnel and gained valuable hands-on experience in interpreting its operational state using Cisco's `show crypto ipsec sa` command. I observed changes in the tunnel status, such as the progression from inactive to active states triggered by interesting traffic, and monitored the number of encrypted and decrypted packets to confirm proper tunnel functionality. This lab not only reinforced my understanding of IPSec VPN concepts but also enhanced my ability to troubleshoot and validate secure connections in real-time, which is a critical skill in network security administration.